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SUPERSEDING
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MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143 LOADING, ASSEMBLING AND PACKING FOR

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Quality Assurance Provisions and Special requirements for the loading, assembling and packing for two types of Guided Missile Safety and Arming Devices designated as: (1) Safety and Arming Device, Guided Missile: M143 - Loaded; (2) Safety and Arming Device, Guided Missile: M143 - Inert (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the folowing specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-D-48175 - Detonator, Electric, PATRIOT Loading, Assembly and Packing for Missile: M143, Mechanical Parts and Assembly for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-63255	- Safety and Arming Device, Guided Missile: M143, Mechanical Parts and Assembly for
MIL-S-63256	- Safety and Arming Device, Guided Missile: M143 Logic Circuit Hybrid for
MIL-S-63335	- Safety and Arming Device, Guided Missile: M143, Electrical Parts and Assemblies for
MIL-W-16878/6	- Wire, Electrical, Type ET, 200°C and 260°C, 250 Volts, (Insulated, High Temperature)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-453	- Radiographic Inspection
MIL-STD-810	- Environmental Test Methods
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

9311100	- Safety and Arming Device, GM: M143, Loaded Assembly
9311070	- Safety and Arming Device, GM: M143, Inert Assembly
9331699	- Packing and Marking for Drums, Metal, Shipping for, Safety and Arming Device, GM: M143

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials, parts and explosive components shall be in accordance with applicable drawings and specifications.

3.2 Assemblies. The complete assemblies shall comply with all requirements specified on Drawing (Dwg.) 9311100 for loaded assemblies and associated drawings and Dwg. 9311070 for inert assemblies and associated drawings and with all requirements specified in the applicable specifications and standards.

3.2.1 Adhesives, lubricants and sealants. Adhesives, lubricants or sealants shall not be used in the assemblies if the shelf life is expired.

3.3 Parts and subassemblies. The electrical and mechanical parts and subassemblies shall comply with all requirements specified in MIL-S-63255, MIL-S-63256 and MIL-S-63335.

3.4 Detonator assembly. The detonator assembly shall comply with all requirements specified in MIL-D-48175.

3.5 Lead assembly (Dwg. 9311077).

3.5.1 Radiographic inspection. Lead assemblies shall be radiographically inspected to assure that no air gaps or voids exist. Radiographic inspection shall be in accordance with MIL-STD-453.

3.5.2 Output. The explosive functioning of the lead assembly shall produce an indentation having an average depth of not less than 0.030 inch when tested as specified in 4.5.10 herein.

3.6 Electro-mechanical (EM) assembly (Dwg. 9311101).

3.6.1 Simulated detonator circuit. Prior to assembly of the electrical assembly to the mechanical assembly, the mechanical assembly electrical resistance between flexible cable conductors E5 and E6 right and left shall be 100± 20 ohms when the shunt wire across the simulated detonator resistor is removed and the detonator rotors are in the armed position.

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3.6.2 Continuity. The electrical resistance between the pins specified in Table I with the S&A Device in any position shall be no greater than 1.0 ohm while applying an open electrical potential no greater than 10.0VDC.

TABLE I - Continuity

J1-14 to J1-7	J1-14 to J1-41
J1-14 to J1-8	J1-14 to J1-44
J1-14 to J1-12	J1-14 to J1-45
J1-14 to J1-13	J1- 6 to J1-34
J1-14 to J1-15	J1-26 to J1-42
J1-14 to J1-16	J1-27 to J1-43
J1-14 to J1-17	J1-30 to J1-46
J1-14 to J1-24	J1-31 to J1-47
J1-14 to J1-29	J1-32 to J1-48
J1-14 to J1-39	J1-33 to J1-49
J1-14 to J1-40	J1-50 to Frame Ground (Base Plate)

3.6.3 Resistance. The electrical resistance shall be as specified in Table II with the S&A Device in the position indicated, while applying an open electrical potential no greater than 10 VDC.

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TABLE II - Resistance

	IDENTIFICATION	J-1 CONNECTOR PINS (+) to (-)	POSITION AND REQUIREMENT (OHMS) SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
a.	Condition Monitor	35 to 15	90 to 110	90 to 110	.99M to 1.1M
b.	Latch Monitor, Right	3 to 15	90 to 110	.99M to 1.1M	90 to 110
c.	Latch Monitor, Left	5 to 15	90 to 110	.99M to 1.1M	90 to 110
d.	Fuze Fire Signal, Right	26 to 15	46 to 68	46 to 68	46 to 68
e.	Fuze Fire Signal, Left	27 to 15	46 to 68	46 to 68	46 to 68
f.	Rotor Monitor, Left	2 to 15	88K to 132K	88K to 132K	NGT* 2
g.	Rotor Monitor, Right	18 to 15	88K to 132K	88K to 132K	NGT* 2
h.	Power/Destruct to ITL command	30 to 32	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
i.	Power/Destruct to Charge Command	30 to 31	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
j.	Charge command to ITL command	31 to 32	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
k.	Charge command to Power/Destruct	31 to 30	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
l.	Ground to ITL Return	14 to 33	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm

*Not Greater Than (NGT)

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3.6.4 Connector pins to case ground isolation resistance. The electrical isolation resistance between J1-50 to the pins specified in Table III with the S&A Device in any position shall be greater than 20 megohms while applying an open electrical potential no greater than 10.0VDC.

TABLE III - Isolation resistance

J1-1	J1-18	J1-34
J1-2	J1-19	J1-35
J1-3	J1-20	J1-36
J1-4	J1-21	J1-37
J1-5	J1-22	J1-38
J1-6	J1-23	J1-39
J1-7	J1-24	J1-40
J1-8	J1-25	J1-41
J1-9	J1-26	J1-42
J1-10	J1-27	J1-43
J1-11	J1-28	J1-44
J1-12	J1-29	J1-45
J1-13	J1-30	J1-46
J1-14	J1-31	J1-47
J1-15	J1-32	J1-48
J1-16	J1-33	J1-49
J1-17		

3.6.5 Five volt test. With 5.0 + 1.0 VDC applied to 5 Volt DC Test Input (+) J1-1 to (-) J1-15, the voltage across the pins specified in Table IV shall be as follows with the S&A Device in the position indicated.

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TABLE IV - Five volt test

IDENTIFICATION	CONNECTOR PINS To (-) J1-15	SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
a. Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
b. ITL TM	J1-4	4.5 ± 1.0VDC	0 ± 0.5VDC	4.5 ± 1.0VDC
c. Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
d. ARM TM	J1-19	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
e. FIRE TM	J1-20	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
f. Converter TM	J1-37	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
g. 100V Capacitor Monitor, Left	J1-38	NGT 1.5VDC	NGT 1.5VDC	NGT 1.5VDC
h. 100V Capacitor Monitor, Right	J1-36	NGT 1.5VDC	NGT 1.5VDC	NGT 1.5VDC

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3.6.6 Operating input voltage. The S&A Device shall meet all electrical output requirements herein specified for all input voltages greater than or equal to 22.5VDC and less than or equal to 31.5VDC on the Power Destruct (+) J1-30 to (-) J1-15 or the Charge Command (+) J1-31 to (-) J1-15 or both.

3.7 Fire circuit characteristics.

3.7.1 Fire circuit charging (Charge Command). With the S&A Device in the Safe and Unlocked Postion, the fire circuit capacitor voltage shall be 70.0VDC minimum (min) at (+) J1-36 (Right) and (+) J1-38 (Left) to (-) J1-15 at a maximum (max) of 2.50 seconds after 25.2 - 0.5VDC is applied to Charge Command (+) J1-31 to (-) J1-15. The initial voltage on (+) J1-36 and (+) J1-38 to (-) J1-15 shall be less than 1.5VDC prior to start of test.

3.7.1.1 Fire circuit capacitor voltage storage. With the S&A Device in the Safe and Unlocked Position, the fire circuit capacitor voltage shall remain 70.0VDC minimum (min) at (+) J1-36 (Right) and (+) J1-38 (Left) to (-) J1-15 for six (6) seconds min., after the 25.2 - 0.5VDC charging voltage is removed from (+) J1-31 to (-) J1-15.

3.7.2 Fire circuit charging (Power/Destruct). With the S&A Device in the Safe and Unlocked position, the fire circuit capacitor voltage shall be 70.0VDC minimum (min) at (+) J1-36 (Right) and (+) J1-38 (Left) to (-) J1-15 at a maximum of 2.50 seconds after 25.2 - 0.5VDC is applied to (+) J1-30 to (-) J1-15. The initial voltage on (+) J1-36 and (+) J1-38 to (-) J1-15 shall be less than 1.5VDC prior to start of test.

3.7.3 No charging while armed. With the S&A Device in the Armed and Locked position, the voltage on the firing circuit capacitors shall be no greater than 15.0VDC at (+) J1-36 (Right) and (+) J1-38 (Left) to (-) J1-15 after 28.0 $\pm$ 2.8VDC is applied for 6 $\pm$ 4 sec to Charge Command (+) J1-31 to (-) J1-15.

3.7.4 Fire circuit function. The left or right fire circuit is defined as having functioned at the time the Fire TM output changes from 0 $\pm$ .5V to 4.5 $\pm$ 1.0V at (+) J1-20 to (-) J1-15. The left fire circuit has functioned when the left fire circuit capacitor voltage (+) J1-38 to (-) J1-15 has discharged to less than 15.0VDC in no greater than 400 msec after the Fire TM output pulse has been detected. The right fire circuit has functioned when the right fire circuit capacitor voltage (+) J1-36 to (-) J1-15 has discharged to less than 15.0VDC in no greater than 400 msec after the Fire TM output pulse has been detected.

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3.7.4.1 Logic destruct.

3.7.4.1.1 Loss of power after arming. The fire circuits shall function in no less than 10 ms and no greater than 50 ms.

3.7.4.1.2 Fire upon arming after loss of power. The fire circuits shall function in no greater than 500 ms prior to either Rotor Monitor (right) (+) J1-18 to (-) J1-15 or Rotor Monitor (left) (+) J1-2 to (-) J1-15 voltage changing from 4.5 ±1.0 to 0 ± 0.5VDC.

3.7.4.1.3 Power destruct threshold. The fire circuits shall function when the voltage applied to Power/Destruct (+) J1-30 to (-) J1-15 is between 18.0 and 21.0VDC.

3.7.4.2 Fuze fire function.

3.7.4.2.1 No fire signal A. Neither fire circuit shall function when a 5.0 ± 0.25V pulse for a duration of 200 ± 10 microseconds with a rise time no greater than one (1) microsecond is applied to Fuze Fire Signal left (+) J1-27 and Fuze Fire Signal right (+) J1-26 to (-) J1-15. At the completion of the no fire signal, both fire circuit capacitor monitors shall remain 70.0VDC minimum.

3.7.4.2.2 No fire signal B. Neither fire circuit shall function when a 10.0 ± 0.5V pulse for a duration of 2.0 + 0.5 microseconds with a rise time no greater than one (1) microsecond is applied to Fuze Fire Signal left (+) J1-27 and Fuze Fire Signal right (+) J1-26 to (-) J1-15. At the completion of the no fire signal, both fire circuit capacitor monitors shall remain 70.0VDC minimum.

3.7.4.2.3 No fire signal C. Neither fire circuit shall function when a 30.0 + 1.5V pulse for a duration of 0.4 ± 0.05 microseconds with a rise time no greater than 0.08 microseconds is applied to Fuze Fire Signal left (+) J1-27 and Fuze Fire Signal right (+) J1-26 to (-) J1-15. At the completion of the no fire signal, both fire circuit capacitor monitors shall remain 70.0VDC minimum.

3.7.4.2.4 Fuze fire signal. Both fire circuits shall function when a Fuze Fire Signal of 18.0 - 2.0V for a duration of 90 - 10 microseconds with a rise time NGT 20 microseconds is applied to Fuze Fire Signal left (+) J1-27 and then Fuze Fire Signal right (+) J1-26 to (-) J1-15.

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3.8 S&A interface outputs (J-1 Connector).

3.8.1 Power TM threshold. The Power TM monitor at (+) J1-21 to (-) J1-15 shall be 4.5 ± 1.0VDC when the Power/Destruct input at (+) J1-30 to (-) J1-15 is greater than 25.0VDC. The Power TM monitor shall be 0 ± 0.5 when the Power/Destruct input is less than 21.0VDC. This transition shall occur at 23.0 ± 2.0VDC.

3.8.2 Voltage outputs. With the S&A Device in the position indicated in Table V the voltage requirements shall be as specified when the input voltage as indicated is applied to Power/Destruct (+) J1-30 to (-) J1-15 for ten (10) seconds min. prior to measurements.

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TABLE V - Interface outputs (J1)

IDENTIFICATION	INPUT VOLTAGE POWER/DESTRUCT (+) J1-30 CONNECTOR PINS To (-) J1-15	(23.0 - 0.5VDC) SAFE & LOCKED	(23.0 - 0.5VDC) SAFE & UNLOCKED	(31.0 + 0.5VDC) ARMED & LOCKED
a. Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
b. ITL TM	J1-4	4.5 ± 1.0VDC	0 ± 0.5VDC	4.5 ± 1.0VDC
c. Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
d. ARM TM	J1-19	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
e. FIRE TM	J1-20	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
f. 5VDC Monitor	J1-23	4.5 ± 1.0VDC	4.5 ± 1.0VDC	4.5 ± 1.0VDC
g. 100V Capacitor Monitor (Right)	J1-36	70 to 110VDC	70 to 110VDC	80 to 110VDC
h. Converter TM	J1-37	4.5 ± 1.0VDC	4.5 ± 1.0VDC	4.5 ± 1.0VDC
i. 100V Capacitor Monitor (Left)	J1-38	70 to 110VDC	70 to 110VDC	80 to 110VDC

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3.9 Intent to launch (ITL) and arming.

3.9.1 ITL command resistance. The electrical resistance from (+) J1-32 to (-) J1-33 shall be as specified in Table VI while applying an open electrical potential no greater than 10VDC at the temperature indicated.

TABLE VI. ITL resistance

FAHRENHEIT	CELSIUS	MINIMUM (OHMS)	MAXIMUM (OHMS)
a. 70 ± 18	21.1 ± 10.1	6.2	7.5
b. -40 ± 5	-40 ± 2.8	4.9	5.5
c. 150 + 5	66 + 2.8	7.6	8.5

3.9.2 Intent-to-launch (ITL). With the S&A Device in the Safe and Locked position and with 5.0 + 1.0VDC applied to (+) J1-1 to (-) J1-15, the Latch Monitor Right and Latch Monitor Left resistances measured between (+) J1-3 (Right) to (-) J1-15 and (+) J1-5 (Left) to (-) J1-15 shall be 100 ohms ± 10% and the ITL TM voltage (+) J1-4 to (-) J1-15 shall be 4.5 ± 1VDC. The application of a 25.2 - 1.0V pulse for a duration of 180 ± 10 milliseconds to ITL Command (+) J1-48 and (+) J1-32 to (-) J1-49 and (-) J1-33 shall cause the S&A to unlock. This unlocking shall be indicated by Latch Monitor resistances of 1M ohm ± 10% and an ITL TM voltage of 0 ± 0.5VDC. The measurements shall be taken 215 ± 15 milliseconds after the start of the applied pulse.

3.9.3 Detonator rotor arming. Arming is defined as the detonator rotors rotating from the Safe and Unlocked position (see 3.11.2.2) to the Armed position when Rotor Monitor (Right) (+) J1-18 to (-) J1-15 Rotor Monitor (Left) (+) J1-2 to (-) J1-15 and Arm TM (+) J1-19 to (-) J1-15 have changed from 4.5 ± 1VDC to 0 ± 0.5VDC. The S&A Device shall be Armed and Locked (See 3.11.3.2) within 300 milliseconds of Arming.

3.9.3.1 Detonator rotor arming time. With the S&A device in the safe and unlocked position, each detonator rotor shall arm in the times specified in Table VII, when the +Z axis (See Figure 1 & 2) is subjected to 11.9 ± 0.2g's at the temperature indicated.

TABLE VII - Arming time

TEMPERATURE (DEGREES)		TIME (SECONDS)	
FAHRENHEIT	CELSIUS	MINIMUM	MAXIMUM
-40°-5° to 150°+5°	-40°-2.8° to 66°+2.8°	3.0	4.1*

*Units tested at -30°F or lower may arm no greater than 4.2 seconds.

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3.9.4 Non-arming. With the S&A device in the Safe and Unlocked position, neither Rotor Monitor (left) (+) J1-2 to (-) J1-15 nor (right) (+) J1-18 to (-) J1-15 shall change from 4.5 ± 1.0VDC when subjected to a constant acceleration of 4.0 + 0.5g's for a period of 20 seconds min. in the + Z axis (see figures 1 and 2).

3.9.5 Return to safe. With no simulated g force applied, the S&A Device shall return to the Safe and Locked position (see 3.11.1.2) from the Armed and Locked position (see 3.11.3.2) when a Return to Safe pulse of 35.0 ± 1.0V with a duration of 350 ± 150 milliseconds is applied to ITL (+) J1-48 and (+) J1-32 to (-) J1-49 and (-) J1-33. The S&A may be rearmed and resafed or removed from the test fixture (if used) or placed base plate down and subjected to additional return to safe pulses or both. The frequency of application of pulses shall not exceed a rate of 4 pulses per minute.

3.9.6 Safe and locked position test. With the S&A device in the Safe and Locked position, both detonator rotors shall remain Safe and Locked (3.11.1.2.1) when subjected to a constant acceleration of 12 + 0.5g's in the +Z axis (see figures 1 and 2) for a period of 20 seconds minimum.

3.10 Limited environmental tests (LET) for the S&A device.

3.10.1 Random vibration (70°F). The S&A Device, in the Safe and Locked position and with 28 ± 1VDC connected to Power/Destruct (+) J1-30 to (-) J1-15, shall be subjected to this test. The test shall be conducted in accordance with MIL-STD-810, method 514 as applicable. During this test, current shall not exceed 500 milliamps. Apply random vibration along each of the three principal axes (see figure 1) for (1) minute min. at the following vibration levels: (see 6.7).

a. 0.035 g^2/HZ from 100 to 2000 HZ.

b. Attenuated 6 decibel (db)/octave from 50 to 100 Hz.

c. Attenuated 6 decibel (db)/octave from 2000 to 3000 Hz.

d. TOTAL: 9.7 RMSg, 50 to 3000 Hz.

3.10.2 Cold operating. The S&A Device shall meet the arming, non-arming and electrical characteristics herein specified at a temperature of -40 -5°F after being temperature conditioned for 2 hours min. at -40 -5°F.

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3.10.3 Hot operating. The S&A Device in the Safe and Locked position with 28 ± 1VDC connected to Power/Destruct (+) J1-30 to (-) J1-15 shall be temperature conditioned for a min. of two (2) hours at 150 + 5°F (66 + 2.8°C). During this test the current shall not exceed 500 milliamps. The S&A Device shall meet the arming, non-arming and electrical characteristics herein specified at a temperature of 150° + 5°F after this conditioning.

3.10.4 Examination. Upon completion of vibration and temperature testing, the cover assembly shall be removed and there shall be no evidence of broken, deformed or displaced parts, nor evidence of foreign material in the S&A Device (see Figure 3).

3.11 Position conditions.

3.11.1 Safe and locked.

3.11.1.1 Visual. The windows shall be transparent and shall not be damaged to the extent that the position of the rotors are obscured. Indication of a red color or the letter "A" in black on a red background directly under either window shall be evidence of the rotors being armed or partially armed.

3.11.1.2 Electrical.

3.11.1.2.1 Resistance. The electrical resistance for Condition Monitor (+) J1-35 to (-) J1-15, Latch Monitor (Right) (+) J1-3 to (-) J1-15 and Latch Monitor (Left) (+) J1-5 to (-) J1-15 shall be 100 ± 10 ohms.

3.11.1.2.2 Voltage. The voltage outputs for Arm TM (+) J1-19 to (-) J1-15 and ITL TM (+) J1-4 to (-) J1-15 shall be 4.5 ± 1VDC, and Rotor Monitor (Right) (+) J1-18 to (-) J1-15 and Rotor Monitor (Left) (+) J1-2 to (-) J1-15 shall be 4.5 ± 1.0VDC.

3.11.2 Safe and unlocked.

3.11.2.1 Visual. The windows shall be transparent and shall not be damaged to the extent that the position of the rotors is obscured. Indication of a red color or the letter "A" in black on a red background directly under either window shall be evidence of the rotors being armed or partially armed.

3.11.2.2 Electrical.

3.11.2.2.1 Resistance. The electrical resistance for Condition Monitor (+) J1-35 to (-) J1-15 shall be 100 ± 10 ohms, and the Latch Monitor (Right) (+) J1-3 to (-) J1-15 and Latch Monitor (Left) (+) J1-5 to (-) J1-15 shall be .99M to 1.1M ohms.

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3.11.2.2.2 Voltage. The voltage outputs for Arm TM (+) J1-19 to (-) J1-15, Rotor Monitor (Right) (+) J1-18 to (-) J1-15 and Rotor Monitor (Left) (+) J1-2 to (-) J1-15 shall be 4.5 ± 1VDC. ITL TM (+) J1-4 to (-) J1-15 shall be 0 ± 0.5VDC.

3.11.3 Armed and locked.

3.11.3.1 Visual. The windows shall be transparent and shall not be damaged to the extent that the position of the rotors are obscured. Indication of a green color or the letter "S" in white on a green background directly under either window shall be evidence of the rotors being safe or partially armed.

3.11.3.2 Electrical.

3.11.3.2.1 Resistance. The electrical resistance for Condition Monitor (+) J1-35 to (-) J1-15 shall be .99M to 1.1M ohms, and Latch Monitor (Right) (+) J1-3 to (-) J1-15. Latch Monitor (Left) (+) J1-5 to (-) J1-15 shall be 100 ± 10 ohms.

3.11.3.2.2 Voltage. The voltage outputs for Arm TM (+) J1-19 to (-) J1-15, Rotor Monitor (Right) (+) J1-18 to (-) J1-15 and Rotor Monitor (Left) (+) J1-2 to (-) J1-15 shall be 0 ± 0.5VDC. ITL TM (+) J1-4 to (-) J1-15 shall be 4.5 ± 1VDC.

3.12 Environmental.

3.12.1 Operating temperature. The S&A device shall meet the functioning, arming and non-arming requirements and electrical characteristics specified herein, at any temperature condition from -40°F to + 150°F. The operating temperature conditions shall be: (See 4.6.1)

a. COLD: -40° - 5°F (-40° - 2.7°C)
b. AMBIENT (AMB): 70° ± 18°F (21° ± 10°C)
c. HOT: 150° + 5°F (66° + 2.8 °C)

Unless otherwise indicated, the operating temperature shall be ambient.

3.12.2 Transportation vibration (T.V.). The Safe and Locked S&A Device shall not explosively function or arm during the T.V. test. After the T.V. test, the S&A Device shall have remained Safe and Locked and shall meet the functioning, arming and non-arming requirements and electrical characteristics herein specified. The Safe and Locked S&A Device shall be subjected to a sinusoidal vibration from 5.5 to 500 to 5.5 Hertz (HZ) for a minimum of four (4) 15 minute Log sweeps, one hour total along each of the three perpendicular axes at the following vibrator accelerations (see 6.7):

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a. 1.0 inch double amplitude from 5.5 to 7.0 Hz.

b. Plus or minus (±) 2.5 g peak from 7.0 to 38 Hz.

c. 0.036 inch double amplitude from 38 to 52 Hz.

d. Plus or minus (±) 5.0 g peak from 52 to 500 Hz.

3.12.3 Launch shock (LS). The Safe and Unlocked S&A Device shall not lock, function or arm during the LS test. After the LS test, the S&A Device shall be Safe and Unlocked and shall meet the functioning, arming and non-arming requirements and electrical characteristics herein specified. The test shall be conducted in accordance with MIL-STD-810, Method 516, Procedure IV, except that the peak value shall be a terminal sawtooth of 172g's and the duration shall be 1.1 milliseconds with the tolerances specified in MIL-STD-810. The pulses shall be applied in each direction of the 3 principal axes (see figure 1) for a total of 6 pulses.

3.12.4 Lateral arming. When a Safe and Unlocked S&A Device with 5.0 + 1.0VDC applied to 5V Test (+) J1-1 to (-) J1-15, is subjected to the acceleration environment specified below, the Detonator Rotors shall arm as specified in Para. 3.9.3. Both Detonator Rotors shall be Armed and Locked upon removal of the acceleration.

a. A force normal to the longitudinal axis (+Z; -Z) of the S&A Device which is equal to or greater than the lateral acceleration force profile in Drawing 9357707.

b. A force along the longitudinal axis (+Z; -Z) of the S&A which is equal to or less than the axial acceleration force shown in Drawing 9357707.

3.12.5 Acceleration (ACCL). The S&A Device shall be Armed and Locked prior to ACCL testing and shall remain Armed and Locked at completion of test. Apply a constant acceleration for three (3) minutes minimum in each of the following axes (see Figure 1):

a. Vector sum of:

46g longitudinal aft-to-forward (- Z) and
52g lateral (- Y)
70g at 48° from longitudinal axis.

b. 22g longitudinal forward-to-aft (+ Z) and
52g lateral (- Y)
57g at 67° from longitudinal axis.

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3.12.6 Flight shock (FS). The S&A Device shall be in the Armed and Locked position during this test and shall remain Armed and Locked at the completion of test. Apply one (1) pulse in each direction of the three principal axes (see Figure 1) for a total of 6 pulses. The pulse shall be a 50g terminal sawtooth having a duration of six (6) milliseconds with the tolerences specified in MIL-STD-810, Method 516, Procedure IV.

3.12.7 Flight vibration (F.V.). The Armed and Locked S&A Device shall not explosively function or unlock during the F.V. test. After the F.V. test, the S&A Device shall have remained Armed and Locked and shall meet the functioning and electrical characteristics herein specified. The test shall be conducted in accordance with MIL-STD-810, method 514 as applicable. The Armed and Locked S&A Device shall be subjected to a random vibration along each of the three principal axes for three (3) minutes min. at the following vibration levels (see 6.7):

a. 0.035 g^2/HZ from 100 to 2000 Hz.

b. Attenuated 6 decibel (db)/octave from 50 to 100 Hz.

c. Attenuated 6 decibel (db)/octave from 2000 to 3000 Hz.

d. TOTAL: 9.7 RMSg, 50 to 3000 Hz.

3.13 Inert safety and arming device (Dwg. 9311070).

3.13.1 General. The Inert S&A Device shall comply with all requirements specified for the S&A Device (see paragraph 3.0).

3.14 Loaded S&A device requirements (Dwg. 9311100).

3.14.1 Continuity. Shall be as specified in 3.6.2.

3.14.2 Resistance. Shall be as specified in 3.6.3.

3.14.3 Connector pin to case ground isolation resistance. Shall be as specified in 3.6.4.

3.14.4 Five volt test. Shall be as specified in 3.6.5.

3.14.5 ITL command resistance. Shall be as specified in 3.9.1a.

3.14.6 Intent-to-launch (ITL). Shall be as specified in 3.9.2.

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3.14.7 Detonator rotor arming. Shall be as specified in 3.9.3.

3.14.8 Non-arming. Shall be as specified in 3.9.4.

3.14.9 Return to safe. Shall be as specified in 3.9.5.

3.14.10 Safe and locked position test. Shall be as specified in 3.9.6.

3.14.11 Fire circuit charging (charge command). With the S&A Device in the Safe and Locked position, the fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+) J1-36 (right) and (+) J1-38 left to (-) J1-15 at a maximum (max.) of 2.60 seconds after 25.2 -0.5VDC is applied to Charge Command (+) J1-31 to (-) J1-15. The initial voltage of (+) J1-36 and (+) J1-38 to (-) J1-15 shall be less than 1.5VDC prior to start of test.

3.14.12 Fire circuit capacitor voltage storage. With the S&A Device in the Safe and Locked position, the fire circuit capacitor voltage shall remain 70.0VDC minimum (min.) at (+) J1-36 (right) and (+) J1-38 (left) to (-) J1-15 for six (6) seconds min., after the 25.2 - 0.5VDC charging voltage is removed from charge command J1-31 to (-) J1-15.

3.14.13 Voltage outputs. With the S&A Device in the Safe and Locked position the voltage requirement shall be as specified in Table VIII, when 28.0 ± 2.0VDC is applied to Power/Destruct (+) J1-30 to (-) J1-15 for ten (10) seconds min. prior to measurements.

TABLE VIII. Outputs (J1)

	IDENTIFICATION	CONNECTOR PINS To (-) J1-15	SAFE & LOCKED
a.	Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC
b.	ITL TM	J1-4	4.5 ± 1.0VDC
c.	Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC
d.	ARM TM	J1-19	4.5 ± 1.0VDC
e.	FIRE TM	J1-20	0 ± 0.5VDC
f.	5VDC Monitor	J1-23	4.5 ± 1.0VDC
g.	100V Capacitor Monitor (Right)	J1-36	80 to 110VDC
h.	Converter TM	J1-37	4.5 to ± 1.0
i.	100V Capacitor Monitor (Left)	J1-38	80 to 110VDC

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3.14.14 Fire circuit stabilized voltage. With the S&A Device in the Safe and Locked position, the Fire Circuit Capacitor Voltage shall be 80.0VDC minimum at (+) J-36 (Right) to (-) J-15 and (+) J-38 (Left) to (-) J1-15 after a minimum of 10 seconds charging with 28 ± 2.8VDC applied at Charge Command (+) J1-31 to (-) J1-15 or Power/Destruct (+) J1-30 to (-) J1-15.

3.14.15 Fire circuit discharging. With the S&A Device in the Safe and Locked position, the fire circuit voltage shall be 15.0VDC max. at (+) J1-36 (right) and (+) J1-38 (left) to (-) J1-15 in 15 minutes max., after 31.0 - 0.2VDC is applied to charge command (+) J1-31 to (-) J1-15 for a minimum of 10 seconds then removed.

3.14.16 Detonator shorting devices. Shorting devices added to the detonator rotors at any time during assembly or testing shall be removed prior to sealing.

3.14.17 Resistance through detonator circuit. The electrical resistance through the detonator shall be one (1) ohm min., and 1.7 ohms max. with the detonator rotor in the Armed and Locked position at 70 ± 18°F (21 ± 10°C) and measured across E1 to E5 on each fire board assembly.

3.14.18 Explosive functioning. With the S&A Device in the Armed and Locked position it shall be considered explosively functioned when the fire circuit function (see 3.7.4) has occurred and caused at least one lead to function high order as specified in 3.5.2.

3.14.18.1 Fuze fire. The S&A Device shall explosively function in less than 200 microseconds on the first attempt, when fuze fire signal inputs (right) (+) J1-26 to (-) J1-15 and fuze fire signal input (left) (+) J1-27 to (-) J1-15 are applied as specified in 3.7.4.2.4. These pulses are to be applied simultaneously.

3.14.18.2 Loss of power. The S&A Device shall explosively function in no greater than ten (10) seconds upon removal of Power/Destruct voltage (+) J1-30 to (-) J1-15.

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3.15 Quality rejection reporting. When failures occur during assembly testing, the testing shall stop at the end of a test sequence which will not compromise that portion of the test. The test conductor shall initiate the Quality Rejection Report as specified in Figure 4 or equivalent.

3.16 Traceability documentation and performance data. The manufacturer shall establish a Traceability Documentation and Performance Data Program in accordance with the requirements of Appendix A.

3.16.1 Test data. Test data from tests specified in 4.3.4, 4.4.2.1, 4.4.2.11, 4.4.2.15, 4.4.2.18, 4.4.2.19, 4.4.2.20, 4.4.2.21, 4.4.2.22, 4.4.2.23, 4.4.2.24, 4.4.2.25, 4.4.2.26 and 4.4.2.29 shall be submitted when requested by the procuring activity. Variable data shall be submitted for those requirements specified in 3.5.2, 3.6.3, 3.6.5, 3.7.1, 3.7.2, 3.7.4.1.1, 3.7.4.1.3, 3.8, 3.9.1, 3.12.4, 3.14.11, 3.14.12, 3.14.13, 3.14.14, 3.14.17 and 3.14.18.1. Attribute data (go, no-go) may be submitted for all other requirements.

3.17 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.18 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, unblended radii, cold solder joints, surface defects, cracks, grease, dirt, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

	Part Description	Drawings	Quantity
a.	Detonator, Electric, PATRIOT	9279861	20
b.	Loaded Shorting Plate Detonator Housing Assembly	9311028	20
c.	Cover Assembly	9311071	6
d.	Cover	9311072	6
e.	Window	9311073	6
f.	Lug, Mounting	9311074	6
g.	Lead Housing Assembly	9311075	6
h.	Lead Housing	9311076	6
i.	Lead Assembly	9311077	20
j.	Cup	9311078	6
k.	Closing Disc	9311079	6
l.	Safety and Arming Device M143, Loaded	9311100	10
m.	Electrical Assembly	9311357	6
n.	Shorting Plate Housing Assy	9321566	6
o.	Packing and Marking for Drums, Metal, Shipping for Safety and Arming Device, GM: M143	9311699	6

4.3.1.1 Homogeneity. The first article shall comply with the homogeneity criteria on MIL-A-48078, Paragraph 4.4.1.

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4.3.2 Inspections to be performed. The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspections of Table IX. All inspections listed in Table IX shall be classified Major unless otherwise specified.

4.3.3 Rejection. See MIL-A-48078.

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TABLE IX. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.3.4	Safety and Arming Device, GM: M143				See 4.3.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Assemblies, Subassemblies and Components (Drawing list - See 4.3.1)				
	Examination for defects	See 4.3.1		3.	4.4.2.1 through 4.4.2.29
	Additional tests to be performed Safety and Arming Device, GM, M143	10 H A C O M O T B L D 4 2 4		3.	4.4.2.23

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formations. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Safety and Arming Device: M143 shall contain:

a. Detonators bearing the same lot interfix number from one manufacturer.

b. RDX from one batch or blending of batches which results in homogeneity of all RDX used in loading the entire S&A lot.

c. Capacitors from not more than two procurement lots from one manufacturer.

d. Silicon controlled rectifiers (SCR's) from not more than two procurement lots from one manufacturer.

e. Hybrids from not more than one procurement lot from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Detonator, Electric, Patriot	1 OF 1	9279861 & MIL-D-48175
			NEXT HIGHER ASSEMBLY: 9311100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bridge resistance (1.0 to 1.5 ohms)		0.40%	3.4	Test
102	Evidence of visual deterioration		0.40%	3.4	Visual
103	Evidence of damage		0.40%	3.4	Visual

NOTES:

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QUALITY ASSURANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Shorting Plate Housing Assembly	1 OF 1	9321566
			NEXT HIGHER ASSEMBLY 9311028

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Excess adhesive		0.40%	3.2	Visual
102	Shorting plate not against detonator housing		0.40%	3.2	Visual
103	Spring pin missing or not below flush		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Loaded Shorting Plate Detonator Housing Assembly	1	1	9311028
				NEXT HIGHER ASSEMBLY: 9311100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Detonator not flush after assembly		0.40%	3.2	Visual
102	Insulating sleeve missing or damaged		0.40%	3.2	Visual
103	Solder not in accordance with MIL-STD-1460		0.40%	3.2	Visual
104	Evidence of solder flow into contact area		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY ASSURANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311028
4.4.2.4	Window				NEXT HIGHER ASSEMBLY 9311071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of body		0.40%	3.2	Gage
102	Height		0.40%	3.2	Gage
103	Improper finish		0.40%	3.2	Visual
104	Not transparent		0.40%	3.10.4.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cover	1 OF 1	9311072
			NEXT HIGHER ASSEMBLY: 9311071

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width (2 places)		0.40%	3.2	Gage
102	Height, max.		0.40%	3.2	Gage
103	Location of cutout for connector		0.40%	3.2	Gage
104	Length of cutout for connector		0.40%	3.2	Gage
105	Height of cutout for connector		0.40%	3.2	Gage
106	Thickness, min.		0.40%	3.2	Gage
107	Location of holes		0.40%	3.2	Gage
108	Diameter of holes		0.40%	3.2	Gage
Minor					
201	Concavity of top, max.		0.65%	3.2	Gage
202	Concavity of sides, max.		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311074
4.4.2.6	Lug, Mounting				NEXT HIGHER ASSEMBLY 9311071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Pitch diameter		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Cover Assembly	1 OF 1	9311071
			NEXT HIGHER ASSEMBLY: 9311101

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of lug from bottom of cover (4 places)		0.40%	3.2	Gage
102	Location of holes (4 places)		0.40%	3.2	Gage
Minor					
201	Weld missing or incomplete		0.65%	3.2	Visual
202	Protective finish missing or damaged		0.65%	3.2	Visual
203	Adhesive missing or excessive around windows		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311076
4.4.2.8	Housing Lead				NEXT HIGHER ASSEMBLY 9311075
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of mounting hole/slot		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Chamfer missing		0.40%	3.2	Gage
104	Diameter of lead hole		0.40%	3.2	Gage
Minor					
201	Thickness of lip at bottom of		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Cup	1 OF 1	9311078
			NEXT HIGHER ASSEMBLY: 9311077

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness at end		0.40%	3.2	Gage
Minor					
201	Length		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Inside diameter		0.65%	3.2	Gage
204	Diameter and concentricity of end, tool only		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311079
4.4.2.10	Disc, Closing				NEXT HIGHER ASSEMBLY 9311077
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Thickness		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311077
4.4.2.11	Lead Assembly				NEXT HIGHER ASSEMBLY 9311075
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of cracks at crimp		0.40%	3.2	Visual
102	Missing or excessive green lacquer		0.40%	3.2	Visual
103	Length		0.40%	3.2	Gage
104	Diameter, max.		0.40%	3.2	Gage
105	Radiographic inspection		0.40%	3.5.1	MIL-STD-453
106	Output	1/	2/	3.5.2	4.5.13.1/Test
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:
1/ A sample size of 20 shall be used for lots of less than 200.
2/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Lead Housing Assembly	1 OF 1	9311075
			NEXT HIGHER ASSEMBLY: 9311100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Green end on wrong end		0.40%	3.2	Visual
102	Excessive adhesive		0.40%	3.2	Visual
103	Lead assembly above flush		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311101
4.4.2.13	Electro-Mechanical Assembly (Visual Prior to Installing Cover Assembly)				NEXT HIGHER ASSEMBLY 9311070/9311100
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fire board mounting screws missing or loose (8 places)		0.40%	3.2	Visual/Manual
102	Logic board mounting screws missing or loose (4 places)		0.40%	3.2	Visual/Manual
103	Conformal coating loose or flaking		0.40%	3.18	Visual
104	Solder not in accordance with MIL-STD-1460		0.40%	3.2	Visual/MIL-STD-1460
105	Solenoid wires not under Flex Cable (2 places)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321571
4.4.2.14	Mounting Plate (Front)				NEXT HIGHER ASSEMBLY 9311101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of two mounting holes		0.40%	3.2	Gage
102	Diameter of holes		0.40%	3.2	Gage
103	Length and width of cutout		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.15	Safety and Arming Device: M143 (Loaded) After Installing Detonator Assembly	1	1	9311100
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Detonator holder screws loose or missing (4 places)		0.40%	3.2	Visual/Manual
102	Insulation damaged		0.40%	3.2	Visual
103	Detonator resistance to fire board assembly		100%	3.14.17	Test
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.16	Safety and Arming Device: M143 (Loaded) After Installing Lead Assembly		1 1		9311100
					NEXT HIGHER ASSEMBLY: N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Lead assembly mounting screws missing or loose		0.40%	3.2	Visual/Manual
Minor 201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.17	Safety and Arming Device: M143 (Loaded) Prior to Installing Cover Assembly	1	1	9311100
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rotors do not rotate from safe to armed position		0.40%	3.2	Manual
102	Rotors not in safe position		0.40%	3.2	Visual
103	Conformal coating loose or flaking		0.40%	3.18	Visual
104	Detonator shorting devices not removed		0.40%	3.14.16	Visual
105	Evidence of foreign material		0.40%	3.18	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.18	Electro-Mechanical Assembly (Visual Prior to Logic and Fire Board Assembly)		1 1		9311101
					NEXT HIGHER ASSEMBLY 9311070/9311100
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Shunt wire not removed from resistor shorting plate detonator circuit (2 places)	100%	1/	3.6.1	Visual/Test

NOTES:
1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.19	Electro-Mechanical Assembly Electrical/Mechanical Operation		1 1		9311101
					NEXT HIGHER ASSEMBLY
					9311100/9311070
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical	None defined				
Major					
102	Continuity	100%	1/	3.6.2	Test
103	Resistance (Safe & Locked)	100%		3.6.3	Test
104	Connector Pins to Case Ground Isolation Resistance	100%		3.6.4	Test
105	ITL Command Resistance	100%		3.9.1	Test
106	ITL Command	100%		3.9.2	Test
107	Fire Circuit Charging (Charge Command)	100%		3.7.1	Test
108	Fire Circuit Capacitor Voltage Storage	100%		3.7.1.1	Test
109	Detonator Rotor Arming	100%		3.9.3	4.5.7/Test
110	Fuze Fire Function	100%		3.7.4.2	4.5.5/Test
111	No Charging While Armed	100%		3.7.3	Test
112	Return to Safe	100%		3.9.5	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.20	Electro-Mechanical Assembly Electrical/Mechanical Operation	1	1	9311101
				NEXT HIGHER ASSEMBLY
				9311100/9311070

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical	None defined				
Major					
113	ITL Command	100%		3.9.2	Test
114	Fire Circuit Charging (Power/Destruct)	100%	1/	3.7.2	Test
115	Fire Upon Arming After Loss of Power	100%		3.7.4.1.2/ 3.9.3	4.5.3/4.5.7/Test
116	Return to Safe	100%		3.9.5	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.21	Electro-Mechanical Assembly Electrical/Mechanical Operation	1 1	9311101
			NEXT HIGHER ASSEMBLY
			9311100/9311070

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical	None defined				
Major					
117	Voltage Output (Safe and Locked)	100%	1/	3.8.2	Test
118	ITL Command	100%		3.9.2	Test
119	Voltage Output (Safe and Unlocked)	100%		3.8.2	Test
120	Detonator Rotor Arming	100%		3.9.3	4.5.7/Test
121	Voltage Output (Armed and Locked)	100%		3.8.2	Test
122	Loss of Power After Arming	100%		3.7.4.1.1	4.5.2/Test
123	Return to Safe	100%		3.9.5	Test
124	Five Volt Test (Safe and Locked)	100%		3.6.5	Test
125	ITL Command	100%		3.9.2	Test
126	Resistance (Safe and Unlocked)	100%		3.6.3	Test
127	Five Volt Test (Safe and Unlocked)	100%		3.6.5	Test
128	Non-arming	100%		3.9.4	4.5.8/Test
129	Power TM Threshold	100%		3.8.1	4.5.6/Test
130	Power/Destruct Threshold	100%		3.7.4.1.3	4.5.4/Test
131	Five Volt Test (Armed and Locked)	100%		3.6.5	Test
132	Resistance (Armed and Locked)	100%		3.6.3	Test
133	Return to Safe	100%		3.9.5	Test
134	Safe and Locked Position Test	100%		3.9.6	Test

NOTES:

1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.22	Electro-Mechanical Assembly Limited Environmental Tests (LET)	1	1	9311101
				NEXT HIGHER ASSEMBLY
				9311070/9311100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical	None defined				
Major					
135	Random Vibration at 70°F	100%	1/	3.10.1	Test
136	Cold Operating at -40°F	100%		3.10.2	4.4.2.19 thru .21/Test
137	Hot Operating at 150°F	100%		3.10.3	4.4.2.19 thru .21/Test
138	Examination	100%		3.10.4	Visual/Manual

NOTES:

1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Safety and Arming Device, M143 (Inert-Loaded) Lot Acceptance Test (LAT)	1 OF 2	9352404/9311100
			NEXT HIGHER ASSEMBLY N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS			AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
		HOT	AMB	COLD			
	Inspection Lot Size 0-200 Sample Size 5				1/		
	TEST SEQUENCE REQUIRED (TSR)						
Critical 1	Functioning, arming or unlocking during T.V.	100%				3.12.2	Test
Major	Dwg. 9352404-1						
101	Electrical/Mechanical Operation	2	1	2		3.6	4.4.2.19/.20/.21/Test
102	Transportation Vibration (Safe and Locked)		5			3.12.2	Test
103	Electrical/Mechanical Operation	2 2/	1 2/	2 2/		3.6	4.4.2.21 category 117-127/Test (TSR)
104	Launch Shock (Safe and Unlocked)	2	1	2		3.12.3	Test
105	Electrical/Mechanical Operation	2	1	2		3.6	4.4.2.24/.20/.21(TSR)
106	Lateral Arming (Safe and Unlocked)	2	1	2		3.12.4/ 3.9.3	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.
2/ Temperature condition optional for all samples.

NOTE: Hot - 150 + 5°F
Amb - 70 ± 18°F
Cold - -40 ± 5°F

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Safety and Arming Device, M143 (Inert-Loaded) Lot Acceptance Test (LAT) (Cont'd)	2 OF 2	9352404/9311100
			NEXT HIGHER ASSEMBLY N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major	Dwg 9352404-2 (See 4.5.11.1)				
107	Acceleration (Armed and Locked)	2 1 2	1/	3.12.5	Test
108	Resistance (Armed and Locked)	2 1 2		3.11.3.1/ 3.11.3.2.1	Visual/Test
109	Flight Shock (Armed and Locked)	2 1 2		3.12.6	Test
110	Resistance (Armed and Locked)	2 1 2		3.11.3.1/ 3.11.3.2.1	Visual/Test
111	Flight Vibration (Armed and Locked)	2 1 2		3.12.7	Test
112	Resistance (Armed and Locked)	2 1 2		3.11.3.1/ 3.11.3.2.1	Visual/Test
	Dwg. 9311100 (See 4.5.11.1)				
113	Explosive functioning (see 4.5.12)	2 0 2 4/ 3/ 4/		3.14.18	Test

NOTES: 3/ Retain for reference (down load to inert).
4/ 50% of sample explosively functioned by loss of power.
50% of sample explosively functioned by application of fuze fire signal.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.24	Safety and Arming Device Electrical/Mechanical Operation (Post Launch Shock)	1 1	9352404-1
			NEXT HIGHER ASSEMBLY 9352404-2

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS			AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
		HOT	AMB	COLD			
	TEST SEQUENCE REQUIRED						
Critical							
1	Functioning, arming or locking during launch shock	100%			1/	3.12.3	
Major							
101	Resistance (Safe and Unlocked)	2	1	2		3.6.3	Test
102	Continuity	2	1	2		3.6.2	Test
103	Connector Pins to Case Ground Isolation Resistance	2	1	2		3.6.4	Test
104	ITL Command Resistance	2	1	2		3.9.1	Test
105	Fire Circuit (Charging Charge Command)	2	1	2		3.7.1	Test
106	Fire Circuit Capacitor Voltage Storage	2	1	2		3.7.1.1	Test
107	Detonator Rotor Arming	2	1	2		3.9.3	4.5.7/Test
108	Fuze Fire Function	2	1	2		3.7.4.2	4.5.5/Test
109	No Charging While Armed	2	1	2		3.7.3	Test
110	Return to Safe	2	1	2		3.9.5	Test

NOTES:

1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Safety and Arming Device; M143 (Loaded) Electrical/Mechanical Operation	1 OF 1	9311100
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical					
1	Visual (Safe and Locked)	100%	1/	3.11.1.1	Visual
Major					
101	5 Volt Test (Safe and Locked)	100%		3.6.5	Test
102	Continuity	100%		3.6.2	Test
103	Resistance (Safe and Locked)	100%		3.6.3	Test
104	Connector Pins to Case Ground Isolation Resistance	100%		3.6.4	Test
105	ITL Command Resistance	100%		3.9.1a	Test
106	ITL Command	100%		3.9.2	Test
107	Resistance (Safe and Unlocked)	100%		3.6.3	Test
108	5 Volt Test (Safe and Unlocked)	100%		3.6.5	Test
109	Detonator Rotor Arming	100%		3.9.3	4.5.9.1/Test
110	5 Volt Test (Armed and Locked)	100%		3.6.5	Test
111	Resistance (Armed and Locked)	100%		3.6.3	Test
112	Return to Safe	100%		3.9.5	Test
113	Non-arming	100%		3.9.4	4.5.9.2/Test
114	Return to Safe	100%		3.9.5	Test
115	Safe and Locked Position Test	100%		3.9.6	Test

NOTES:
1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.26	Safety and Arming Device; M143 (Loaded) Electrical Operation, CRAF Test Set	1	1	9311100
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical					
1	Visual Safety (Safe & Locked)	100%		3.11.1.1	Visual
Major					
101	Continuity	100%		3.6.2	Test
102	Resistance	100%		3.6.3	Test
103	Connector Pins to Case Ground Isolation Resistance	100%	1/	3.6.4	Test
104	Voltage Outputs (Safe and Locked)	100%		3.14.13	Test
105	Fire Circuit Stabilized Voltage	100%		3.14.14	Test
106	Fire Circuit Discharging	100%		3.14.15	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Safety and Arming Device: M143 (Loaded) Visual Prior to Packing	1 OF 1	9311100
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Visual Safety (Safe and Locked)		100%	3.11.1.1	Visual
Major					
101	Cover Mounting Screws Not Flush or Below on Base Plate		100%	3.2	Visual
102	Damaged or Scratched Windows		100%	3.18	Visual
103	Identification and Marking Incorrect or Illegible		100%	3.2	Visual
104	Damaged or Bent Connector Pins		100%	3.18	Visual
105	Evidence of Dents in cover or base Plate		100%	3.2	Visual
106	Foreign Material in Connector Screw Threads (2 Places)		100%	3.18	Visual
107	Foreign material inside connector or on pins		100%	3.18	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.18	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.28	Metal Shipping Drum (Prior to Sealing)		1 1	9331699
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%		Visual/Manual
102	Missing bag		0.40%		Visual
103	Improper orientation		0.40%		Visual
104	Missing desiccant		0.40%		Visual
105	Vent in cover not properly positioned over cutout in cushion or filler		0.40%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63257B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9331699
4.4.2.29	Metal Shipping Drum				NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Marking incorrect or incomplete		0.40%		Visual
102	Contents loose		0.40%		Manual
103	Missing metallic lead seal		0.40%		Visual
104	Leak test		100%		Test/Dwg 8806652 1/
105	Vent cap not closed		100%		Visual

NOTES:
1/ Only one test (method optional) required per Dwg 8806652.

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4.4.3 Testing.

4.4.3.1 Check test for deterioration of electric detonators (See 6.4). If the total time between original acceptance of the detonator lot and the assembly of the lot into the Safety and Arming Device exceeds five years, or if the detonator packing shows evidence of having been subjected to adverse environmental conditions, however brief, at any time since previous test, the detonator lot shall be subjected to and must pass the check test for possible deterioration specified in MIL-D-48175. This test shall be performed, when required, by the contractor on detonators selected by the inspector at the facility assembling the detonators into Safety and Arming Devices. The test shall be conducted at Government expense without cost to the contractor who produced the detonators or the contractor assembling the detonators into the Safety and Arming Device, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., of the detonator lot while the detonator lot was under the jurisdiction of either contractor.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspectin Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and procesing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections - The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein.

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4.5.2 Procedure for loss of power after ARMING test.

4.5.2.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. S&A Device Safe and Unlocked.

b. 31.0 + 0.5VDC applied to Power/Destruct.

c. Left capacitor monitor @ 80.0VDC min.

d. Right capacitor monitor @ 80.0VDC min.

4.5.2.2 Dynamic conditions.

a. Remove 31.0 + 0.5VDC on Power/Destruct.

b. Measure time (3.7.4.1.1) to function.

4.5.3 Procedure for fire upon arming after loss of power test.

4.5.3.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. S&A Device Safe and Unlocked..

b. 25.2 -0.5VDC on Power/Destruct.

c. +Z axis out of line (0 $\pm$ .2g's).

d. Left capacitor monitor @ 70.0VDC min.

e. Right capacitor monitor @ 70.0VDC min.

4.5.3.2 Dynamic conditions.

a. Remove 25.2 - 0.5VDC on Power/Destruct.

b. Perform Arming as specified in 4.5.7.

c. Measure time (3.9.3.1/3.7.4.1.2) to function.

4.5.4 Procedure for power/destruct threshold test.

4.5.4.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

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a. S&A Device Armed and Locked.

b. 23.0 ± 2.0VDC applied to Power/Destruct.

c. Left Capacitor Monitor @ 70VDC min.

d. Right Capacitor Monitor @ 70VDC min.

4.5.4.2 Dynamic conditions. Decrease Power/Destruct input voltage ((+) J1-30 to (-) J1-15) with a voltage step no greater than 25 millivolt (mv) and record input voltage level at the time the fire circuit functions (see 3.7.4).

4.5.5 Procedure for fuze fire function.

4.5.5.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. S&A Device Armed and Locked.

b. 23.0 - 0.5VDC applied to Power/Destruct.

c. +Z acceleration @ 11.9 ± 0.2g's.

d. Left Capacitor Monitor @ 70.0VDC min.

e. Right Capacitor Monitor @ 70.0VDC min.

4.5.5.2 Dynamic conditions.

a. Apply NO Fire Signal A (3.7.4.2.1)
b. Apply NO Fire Signal B (3.7.4.2.2)
c. Apply NO Fire Signal C (3.7.4.2.3)
d. Apply Fuze Fire Signal (3.7.4.2.4)

4.5.6 Procedure for power TM threshold test.

4.5.6.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. S&A Device Armed and Locked.

b. 28.0 ± 3.0VDC aplied to Power/Destruct.

c. Left Capacitor Monitor @ 70.0VDC min.

d. Right Capacitor Monitor @ 70.0VDC min.

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4.5.6.2 Dynamic conditions. Decrease Power/Destruct input voltage using a voltage step no greater than 25 millivolts (mv) and record input voltage at the time the Power TM output voltage changes state (see 3.8.1).

4.5.7 Procedure for detonator rotor arming test (EM Assy).

4.5.7.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. S&A Device Safe and Unlocked.

b. 23.0 - 0.5VDC applied to Power/Destruct.

c. Left Capacitor Monitor @ 70.0VDC min.

d. Right Capacitor Monitor @ 70.0VDC min.

4.5.7.2 Dynamic conditions. The test equipment shall retain the S&A Device in a position such that its +Z axis initially is subjected to 0 ± 0.2g's while the X or Y axis is being subjected to an acceleration level of 11.9 ± 0.2g's ("Out-of-Line" See figure 2a). When the test acceleration level has been attained, the test equipment shall smoothly transition the +Z axis of the S&A 90 ± 2°, within a period of time not greater than 100msec, to the condition where the S&A +Z axis is now experiencing the test acceleration level ("In-Line" See figure 2b). The S&A Device shall be positioned such that the axis of rotation of the test equipment is through the S&A's center of gravity (CG). Record the times as specified in 3.9.3.1.

4.5.8 Procedure for non-arming test (EM Assy).

4.5.8.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. S&A Device Safe and Unlocked.
b. 28.0 ± 3.0VDC applied to Charge command.
c. Left Capacitor Monitor @ 70.0VDC min.
d. Right Capacitor Monitor @ 70.0VDC min.

4.5.8.2 Dynamic conditions. The test equipment shall preposition the +Z axis of the S&A Device (see figure 1) in the g field (see figure 2b) and subject the +Z axis to a constant 4.0 + 0.5g acceleration. The S&A Device shall not arm as specified in 3.9.4. At the completion of this test the test equipment shall increase the acceleration on the +Z axis to a level that will allow both detonators to Arm and Lock. The test equipment shall not exceed a g level of 12.1g's.

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4.5.9 Loaded assembly.

WARNING: ALL TESTS SHOULD BE PERFORMED BEHIND A SUITABLE BARRICADE WHEN TESTING LOADED S&A DEVICES.

4.5.9.1 Procedure for detonator rotor arming test (loaded).

4.5.9.1.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. S&A Device Safe and Unlocked.
b. 5.0 + 1.0/VDC applied to 5V test.
c. Left Capacitor Monitor @ 1.5VDC max.
d. Right Capacitor Monitor @ 1.5VDC max.

4.5.9.1.2 Dynamic conditions. The test equipment shall retain the S&A Device in a position such that its +Z axis initially is subjected to 0 $\pm$ 0.2g's while the X or Y axis is being subjected to an acceleration level of 11.9 $\pm$ 0.2g's ("Out-of-Line" see figure 2a). When the test acceleration level has been attained, the test equipment shall smoothly transition the +Z axis of the S&A 90 $\pm$ 2°, within a period of time not greater than 100msec, to the condition where the S&A +Z axis is now experiencing the test acceleration level (In-Line: see figure 2b). The S&A Device shall be positioned such that the axis of rotation of the test equipment is through the S&A's center of gravity (CG). Record the times as specified in 3.9.3.1.

4.5.9.2 Procedure for non-arming test (loaded).

4.5.9.2.1 Initial conditions. The following conditions shall exist and be verified prior to the start of test:

a. S&A Device Safe and Unlocked
b. 5.0 + 1.0VDC applied to 5V + DC test
c. Left Capacitor Monitor @ 1.5VDC max.
d. Right Capacitor Monitor @ 1.5VDC max.

4.5.9.2.2 Dynamic conditions. The test equipment shall preposition the +Z axis of the S&A Device (see figure 1) in the g field (see figure 2b) and subject the +Z axis to a constant 4.0 + 0.5g acceleration. The S&A Device shall not arm as specified in 3.9.4. At the completion of this test the test equipment shall increase the acceleration on the +Z axis to a level that will allow both detonators to Arm and Lock. The test equipment shall not exceed a g level of 12.1g's.

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4.5.9.3 Lot acceptance test (LAT) modification. The LAT samples selected for this test configuration require modification prior to environmental testing. With the cover removed, remove the jumper wire from E1A to E1 on each fire board asembly. Then solder (MIL-STD-1460 applies) a jumper wire (MIL-W-16878/6, Type PTFE MI16878/6 BEE) from E1A to E6 on each fire board assembly. This wire shall be attached with minimum slack. After modification and in the Safe and Locked position, the resistance shall not be less than 1.6M ohms and not greater than 2.5M ohms from (+) J1-41 and (+) J1-44 to (-) J1-15.

4.5.9.4 Detonator resistance to fire board. With the detonator rotors in the Armed and Locked position, the resistance (see 3.14.17) shall be measured between terminal pins E1 to E5 on each Fire Board Assembly for the detonator resistance. The S&A Device shall be classified defective if one or both detonator circuits fails the required resistance measurements.

4.5.9.5 Procedure for Explosive Function Test. The samples selected for this test require adherence to the following procedure prior to start of test. Remove cover assembly (9311071), resafe if necessary. Remove inert lead housing (9311076) and install (live) lead housing assembly (9311075) and manually Arm unit prior to replacing cover assembly (9311071). Confirm unit is Armed, by visually checking that both windows are RED.

4.5.10 Lead assembly.

4.5.10.1 Functioning. The PATRIOT detonator and Lead Assembly shall be assembled as shown in Figure 5. The lead wires shall be connected to the test circuit in Figure 6 and charged to 80 volts minimum.

CAUTION: The detonator and lead assembly shall be placed in a suitable chamber to perform this test. If the lead assembly fails to explode, wait a minimum of 10 minutes before removing the detonator from the safety chamber.

4.5.10.2 Output. The output of the lead assembly shall be determined by indentation in the steel block as specified in MIL-STD-331 Test 301.1 with the exception that the steel block shall be a minimum of 4 inches square. The point of the dial indicator probe shall have a $30^{\circ} \pm 1^{\circ}$ included angle and the end of the point shall have a radius of 0.025 ± 0.002 inch.

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Detonators which fail to explode shall be classified defective in the function test requirement only, and an additional unit from the lot shall be selected to test for output requirements.

CAUTION: The detonator and lead assembly shall be placed in a suitable chamber to perform this test. If the detonator fails to explode, wait a minimum of 10 minutes before removing the detonator from the safety chamber.

4.6 Environmental.

4.6.1 Temperature conditioning.

4.6.1.1 Extreme. For test involving extreme temperature conditions (hot or cold), the S&A Device or EM Assembly shall be temperature conditioned for a minimum of two (2) hours after the temperature chamber has been stabilized. Longitudinal Acceleration (not Lateral Acceleration) testing of temperature conditioned items shall be performed with the S&A Device in a polystyrene foam container with a minimum thickness of one (1) inch on each surface on the S&A Device. Test sequences which require the S&A or EM Assembly to remain outside of the chamber for more than twenty (20) minutes shall be reconditioned to temperature for a minimum of twice the time (not to exceed 2 hours) the item has been outside the chamber.

4.6.1.1.1 Extreme. For tests involving extreme temperature conditions (hot or cold), the subassemblies or components shall be temperature conditioned for a minimum of one (1) hour after the temperature chamber has been stabilized. Test sequences which require the subassemblies or components to remain outside of the chamber for more than twenty (20) minutes shall be reconditioned to temperature for a minimum of twice the time (not to exceed 1 hour) the item has been outside the chamber.

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5. PACKAGING.

5.1 Packing and marking.

5.1.1 Level A. The assemblies shall be packed and marked in accordance with dwg. 9331699.

5.2 Data cards. Data cards shall be prepared for each lot in accordance with 6.1 of MIL-A-48078. Data cards shall include, as a minimum, the following information:

a. Mechanical assembly
b. Electrical assembly
c. Electro-Mechanical assembly
d. Detonator assembly
e. Lead assembly
f. Serial numbers of all loaded or inert assemblies in the lot.

6. NOTES.

6.1 Intended use. It is intended that the item of this specification will be used in the PATRIOT surface-to-air missile.

6.2 Ordering data. See MIL-A-48078.

6.2.1 Contract data requirements. The requirements for submission of data from the requirements paragraphs will be listed directly on a DD Form 1423 incorporated into the contract. Data Item Description DI-T-1906 will be required in connection with this specification

6.2.2 Submission of data. Submit data, as required, to Commander, US Army Armament, Research and Development Center, ATTN: AMSMC-QAR-R (D), Dover, New Jersey 07801-5001.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament, Research and Development Center, ATTN: AMSMC-QAR-I (D), Dover, New Jersey 07801-5001.

6.4 Check test for deterioration of electric detonators. The Contracting Officer shall arrange to have the contractor reimbursed for expense incurred in performing the check test for deterioration when the test is required and performed.

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6.5 Orientation definition (see Figures 1 and 2). The direction of missile flight is in the -z axis and the base plate is the forward end. In the unarmed or safe condition, the "G" weights are in the forward position and the detonator rotors are locked. The longitudinal axis of the assembly is defined as the axis perpendicular to the base plate.

6.6 Calculating revolutions per minute (RPM). The radius of rotation in inches shall be measured to within 1/16 (0.0625) inches from the center of rotation of the centrifuge to the center of the detonator rotor shaft (1.9 + 0.02 inches from the bottom of the base plate). All measurements are with the S&A device lying along a path of radial acceleration and the forward end (base) pointed toward the center of rotation. The swing fixture pivot point shall be located at the detonator rotor shaft. The following equation is used to produce the acceleration forces required at the predetermined radii of rotation.

$$N = \sqrt{\frac{g \times 10^9}{R \times 28416}}$$

Where:

N = Revolutions per minute (RPM)
g = Acceleration level required
R = Radius of rotation in inches (in.)

For any g level with a tolerance, the revolutions per minute shall be calculated for each end of the tolerance zone. The two results shall be considered as the tolerance applicable to the revolutions per minute of the centrifuge.

6.7 Tolerances. Unless otherwise specified, the maximum allowable tolerances on test condition shall be as follows:

a. Vibration amplitude: Sinusoidal, plus or minus (±) 0.01 inch double amplitude or ± 0.2g whichever is less.

b. Vibration frequency: Plus or minus (±) 2 percent, or ± 1/2 HZ below 25 HZ.

c. Power spectral density: Plus or minus (±) 3 decibel (db) between 100 to 2000 HZ.

d. Sweep time: Sinusoidal: 5 minutes
Random: 15 seconds

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6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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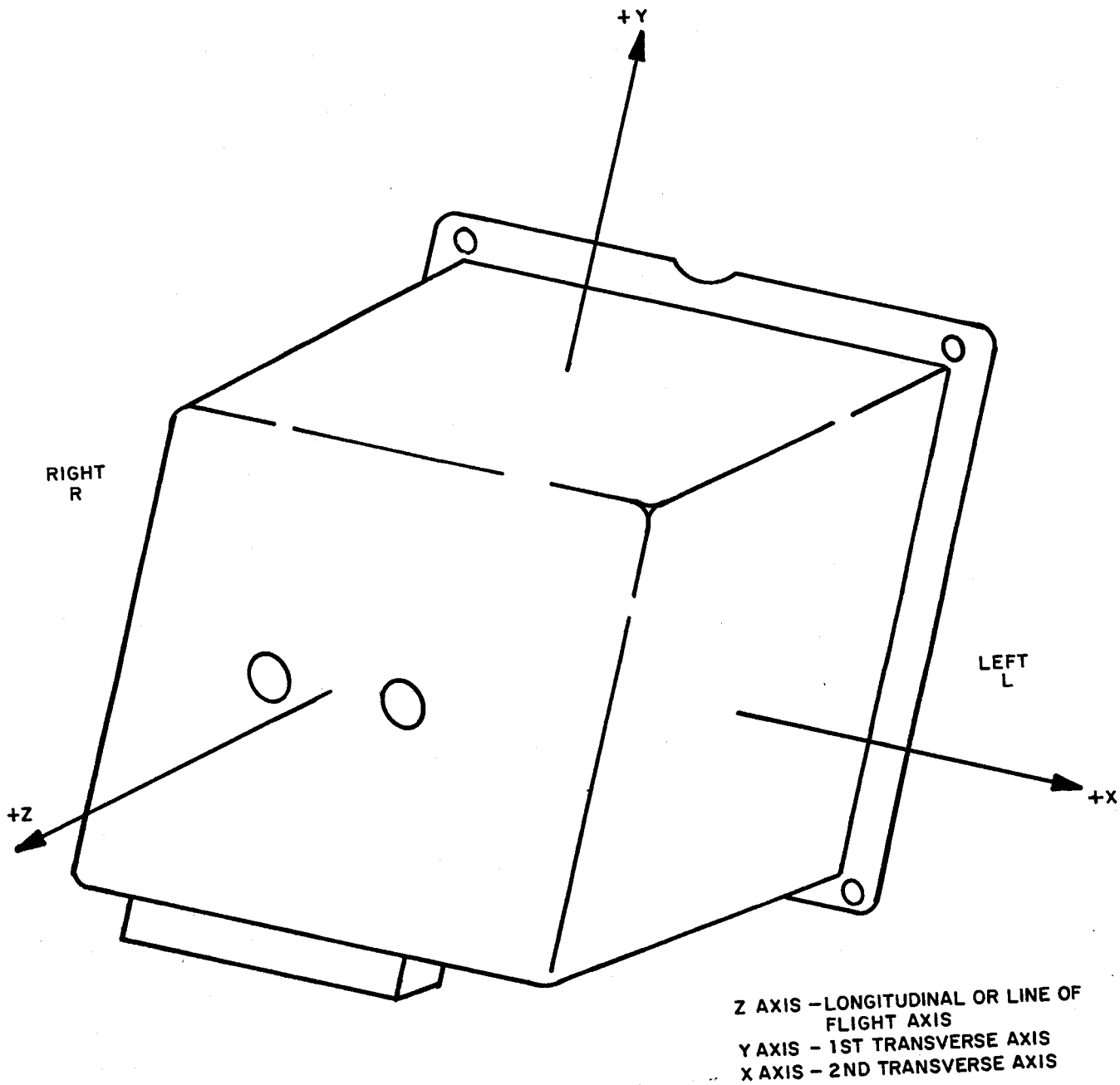


FIGURE 1

PATRIOT S&A DEVICE ORIENTATION

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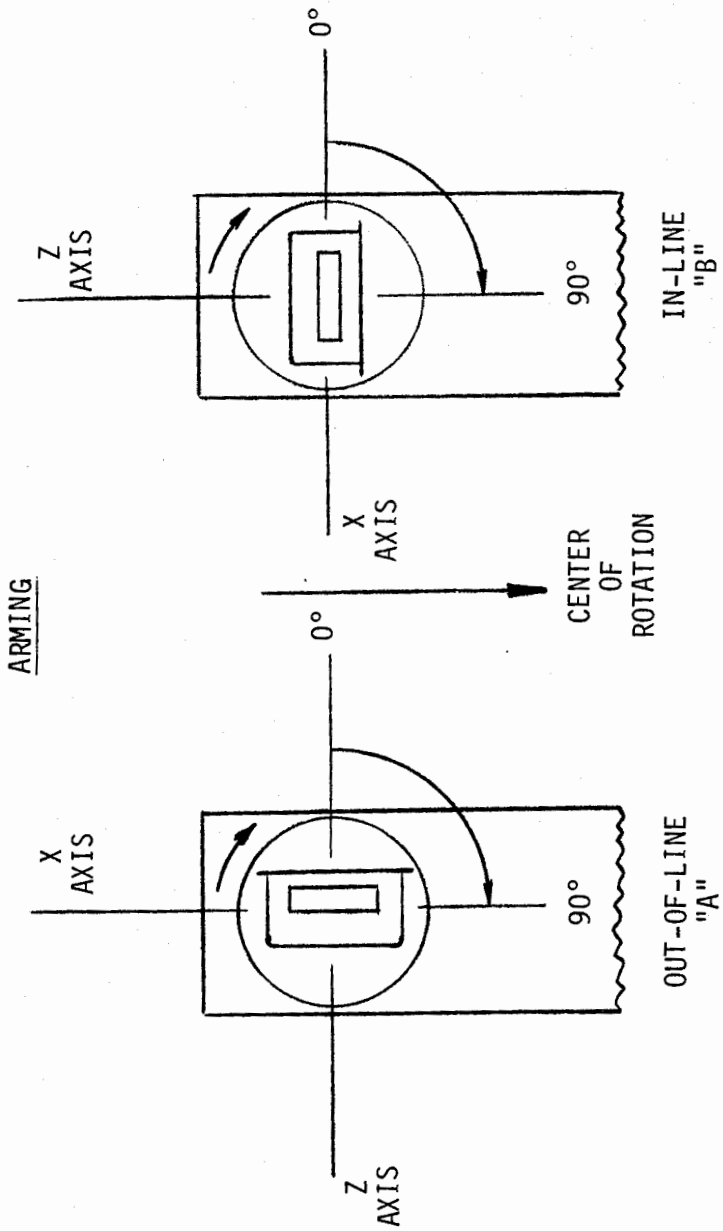


FIGURE 2
Patroit S&A Orientation on Centrifuge Arm

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(RECOMMENDED FORMAT)

PATRIOT

LIMITED ENVIRONMENTAL TESTS (LET)

FOR SAFETY AND ARMING DEVICE:

M143

Serial Number _____ was successfully LET tested according to paragraph 3.10 of MIL-S-63257.

The LET test data are stored in _____. Functional and electrical test data is included in the official S and A device folder.

Examination: Upon completion of the hot temerature operating test the cover assembly shall be removed and there shall be no evidence of broken, deformed or displaced parts, nor evidence of foreign material in the electromechanical assembly. The cover assembly shall be reassembled prior to next assembly test.

Test Data Summary:

	DATE	TESTED BY
Random Vibration at 70°F Completed:		
Storage at -40° Completed:		
Cold Operating @ 40°F		
Storage at 150°F Completed:		
Hot Operating at 150°F:		
Examination:		
Audit:		

FIGURE 3. Limited environmental tests data sheet

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RECOMMENDED FORMAT

PATRIOT

QUALITY REJECTION REPORT

1. a. Part Assy: _____ b. Date Tested: _____

   c. Dwg & Rev: _____ d. SN/PC: _____

   e. Spec & Rev: _____ f. Para. No. ____ Defect No. ____

_____

INSPECTOR or TEST OPERATOR

2. a. Failure Analysis:

   b. Cause of Failure:

   c. Rework Required:

_____

PROJECT ENGINEER DATE

3. a. Rework Required:

   b. Inspection Procedure:

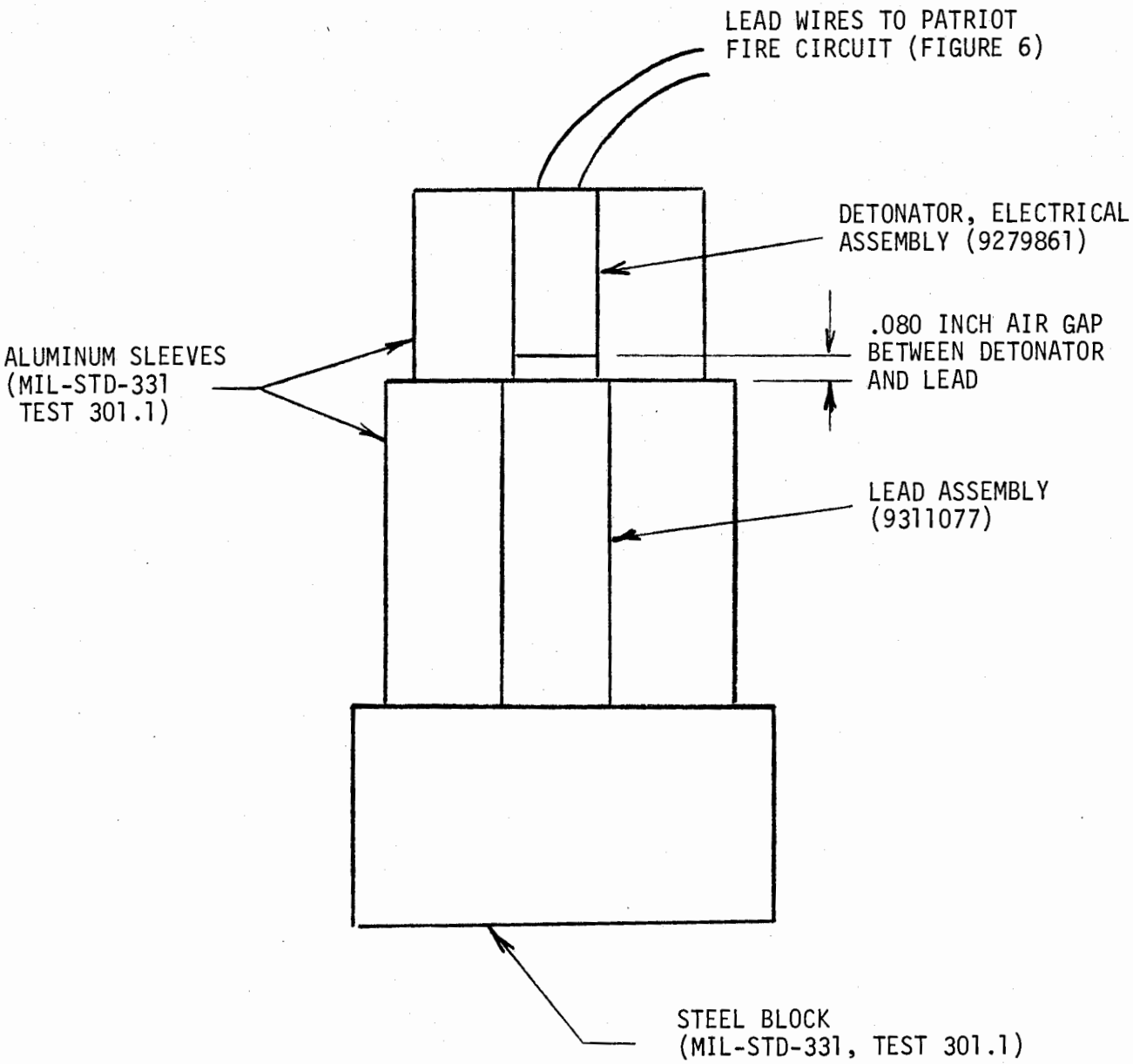
   c. Corrective Action:

_____ _____

QA ENGINEER PROJECT ENGINEER

FIGURE 4. Quality rejection report

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LEAD ASSEMBLY OUTPUT TEST

FIGURE 5

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PARTS LIST: See Fire Board Assembly Apparatus list Dwg. 9311342, Sheet 2.

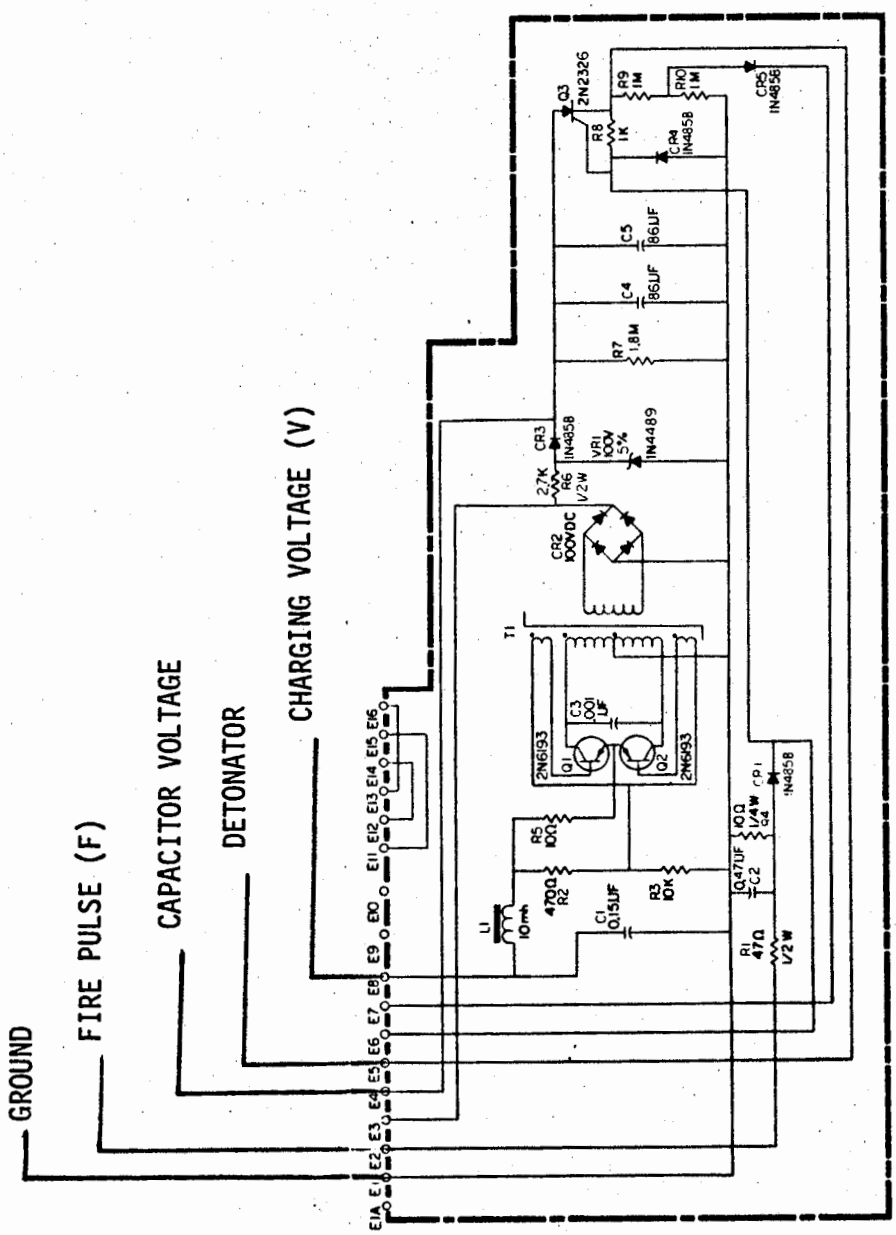
SCHEMATIC: See Fire Board Assembly, Dwg. 9311342, Sheet 3.

	AMPLITUDE		DURATION	
	Minimum	Maximum	Minimum	Maximum
Charging Voltage (Note*)	27.0	20.0	10sec	20sec
Fire Pulse	20.0	28.0	100msec	200msec
Capacitor Voltage	80.0	100.0	3sec	5sec

*Note: Remove on or before Fire Pulse.

Figure 5 (Cont'd). Lead assembly output test

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PATRIOT FIRE CIRCUIT
FIGURE 6

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Custodian:
Army-AR

Preparing activity
Army-AR

(Project: 1336-A498)

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APPENDIX A

TRACEABILITY DOCUMENTATION AND PERFORMANCE DATA

10. SCOPE

10.1 Scope - This appendix describes the trace and performance data required during the manufacture, assembly and testing of the Safety and Arming Device, M143.

20. APPLICABLE DOCUMENTS

20.1 Specifications and standards. The following document of the issue in effect on date of this information for bids or request for proposals form a part of this Appendix to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-63256 - Safety and Arming Device, M143, Logic Circuit Hybrid for

30. Traceability documentation and performance data. Traceability documentation and performance data shall be confirmed by documentation to the Government or to users with government contracts or subcontractors regardless of whether the components or devices are procured directly from the manufacturer or from another source as a distributor. When other sources are involved, their documentation shall be in addition to and include that provided by the manufacturer and previous distributors. This documentation shall include the following informations:

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APPENDIX A

TABLE 1. Traceability and performance

	PART DESCRIPTION	DRAWING	DOCUMENTATION TRACE	DOCUMENTATION PERFORMANCE	DATA REQUIRED MIL-SPEC	DATA REQUIRED PARAGRAPH
a.	S&A Device, Loaded	9311100	REQUIRED	REQUIRED	MIL-S-63257	
b.	Detonator	9279861	REQUIRED	REQUIRED	MIL-D-48175	
c.	Lead Assembly	9311077	REQUIRED	REQUIRED	MIL-S-63257	
d.	Electro-Mechanical	9311101	REQUIRED	REQUIRED	MIL-S-63257	
e.	Mechanical	9311101	REQUIRED	REQUIRED	MIL-S-63255	
f.	Switch Board	9311020	REQUIRED	REQUIRED	NA	NA
g.	Solenoid, Rotor	9311080	REQUIRED	REQUIRED	NA	NA
h.	Switch Subminiature	9311080	REQUIRED	REQUIRED	NA	NA
i.	Plate Shorting, Det.	9311384	REQUIRED	REQUIRED	NA	NA
j.	Plate Shorting, S&A	9311385	REQUIRED	REQUIRED	NA	NA
k.	Electrical Assembly	9311357	REQUIRED	REQUIRED	NA	NA
l.	Logic Circuit Board Assy	9311372	REQUIRED	REQUIRED	MIL-S-63257	
m.	Capacitor, 68 microfarad	9311372	REQUIRED	NA	NA	NA
n.	Logic Circuit Hybrid	9311370	REQUIRED	MIL-S-63256		
o.	Fire Board Assy	9311342	REQUIRED	REQUIRED	MIL-S-63335	
p.	Capacitor, 86 Microfarad	9311342	REQUIRED	NA	NA	NA
q.	Diode, 1N4489	9211342	REQUIRED	NA	NA	NA

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APPENDIX A

TABLE 1. Traceability and performance

	PART DESCRIPTION	DRAWING	DOCUMENTATION		DATA REQUIRED	
			TRACE	PERFORMANCE	MIL-SPEC	PARAGRAPH
r.	Diode, 1N485B	9311342	REQUIRED	NA	NA	NA
s.	SCR, 2N2326 JANTX	9311342	REQUIRED	NA	NA	NA
t.	Choke	9311353	REQUIRED	NA	NA	NA
u.	Transformer	9311354	REQUIRED	NA	NA	NA
v.	Bridge Rectifier	9311356	REQUIRED	NA	NA	NA
w.	Transistor, 2N6193	9311359	REQUIRED	NA	NA	NA

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